

Subject: Application for Thesis in Computer Vision Simulation (f/m/d)

Dear Hiring Team,

I am currently pursuing my Master's degree in Electrical and Information Technology at Hochschule Darmstadt, specializing in Automation and Robotics. I am presently completing a mandatory internship at Bär Automation GmbH in Stuttgart and expect to have completed 80 ECTS credits by May 2026, after which I will be available to begin my Master's thesis. The advertised thesis on computer vision simulation at Volkswagen Commercial Vehicles strongly aligns with my interests, particularly the work on trajectory prediction and simulation-based testing for autonomous driving systems. As someone passionate about robotics and intelligent mobility systems, the opportunity to contribute to technologies developed by Volkswagen one of the leading automotive companies in the world is especially motivating for me.

During my internship as a Mobile Robots Development Intern at Bär Automation GmbH within the ARENA2036 research environment in Stuttgart, I gained practical experience working with autonomous mobile robots and their navigation systems. Using ROS 2 and the Nav2 stack, I explored and compared SLAM approaches such as SLAM Toolbox and Google Cartographer by generating and evaluating maps under different configurations. I also worked on improving robot localization by tuning LiDAR scan matching and odometry parameters to reduce positional drift and improve navigation accuracy. In addition, I became familiar with the VDA 5050 standard used for communication between AGVs and fleet management systems. As part of the same environment, I also contributed to the development of digital twin tools by helping build a Python and React-based tool that automatically generates Asset Administration Shells (AAS v3.1) for industrial assets. This work gave me exposure to structured data models and the integration of physical systems with their digital representations.

In addition to my internship work, I gained experience with robotic perception systems through a university project where my team implemented a mapping and perception pipeline in Python and ROS for a UniTree quadruped robot equipped with a Hesai XT16 LiDAR sensor. The project involved processing point cloud data and generating indoor maps using filtering and segmentation techniques. Through this work, I developed a practical understanding of how robots interpret spatial data and convert it into representations that support navigation and environment understanding.

This experience relates closely to the work described in your thesis position, where accurate perception and environment modeling are essential for testing and validating autonomous systems. Working directly with LiDAR data and mapping pipelines helped me understand challenges such as sensor noise, data preprocessing, and environment representation in both simulated and real-world scenarios.

Before starting my Master's program, I worked as a Senior System Engineer at Infosys. In that role, I worked with enterprise application systems and supported Agile development processes. During a management transition, I also served as a Technical Scrum Master for four months, helping coordinate development activities and supporting the team's workflow.

I would welcome the opportunity to contribute to the research and prototyping work within the Volkswagen Commercial Vehicles team and to develop my Master's thesis in this field. Thank you for your time and consideration.

Best regards,
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